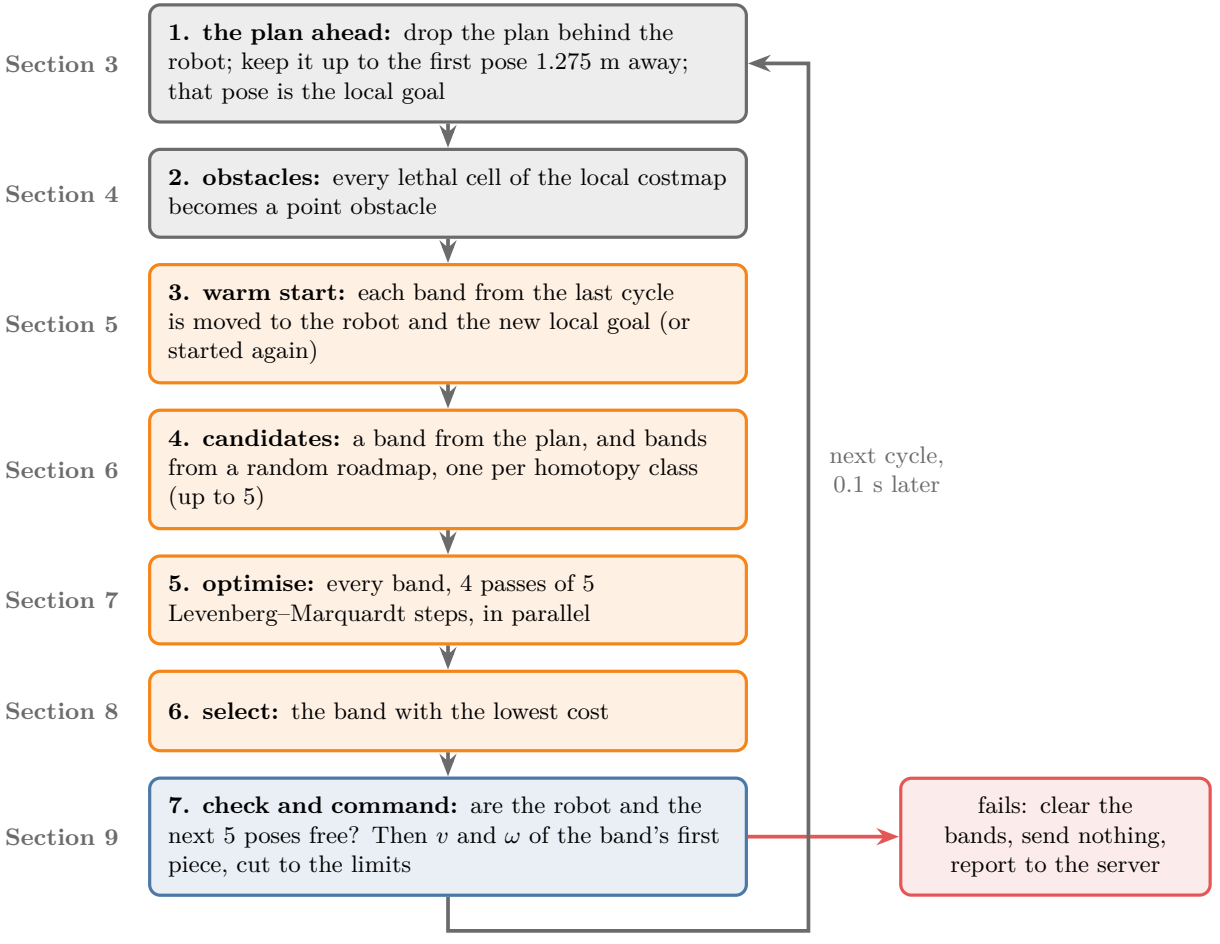

Section 3
1. the plan ahead: drop the plan behind the robot; keep it up to the first pose 1.275 m away; that pose is the local goal
Section 4
2. obstacles: every lethal cell of the local costmap becomes a point obstacle
Section 5
3. warm start: each band from the last cycle is moved to the robot and the new local goal (or started again)
Section 6
4. candidates: a band from the plan, and bands from a random roadmap, one per homotopy class (up to 5)
Section 7
5. optimise: every band, 4 passes of 5 Levenberg–Marquardt steps, in parallel
Section 8
6. select: the band with the lowest cost
Section 9
7. check and command: are the robot and the next 5 poses free? Then v and ω of the band's first piece, cut to the limits
fails: clear the bands, send nothing, report to the server
next cycle, 0.1 s later